



Summary

- 1- Image and video reading**
- 2- Rescaling and resizing a window**
- 3- Drawing shapes and putting text**
- 4- Face detection**
- 5- Face recognition**

Introduction

In the project, OpenCV is used for crop recognition; it is the platform of the project which performs the operations for image and video processing. It will be linked to ROS to control the car and the robot arm.

The goal of this work is to master OpenCV and know how it works; for that, I used a complete dataset (images, videos) on the internet for learning OpenCV.

1- Image and Video reading

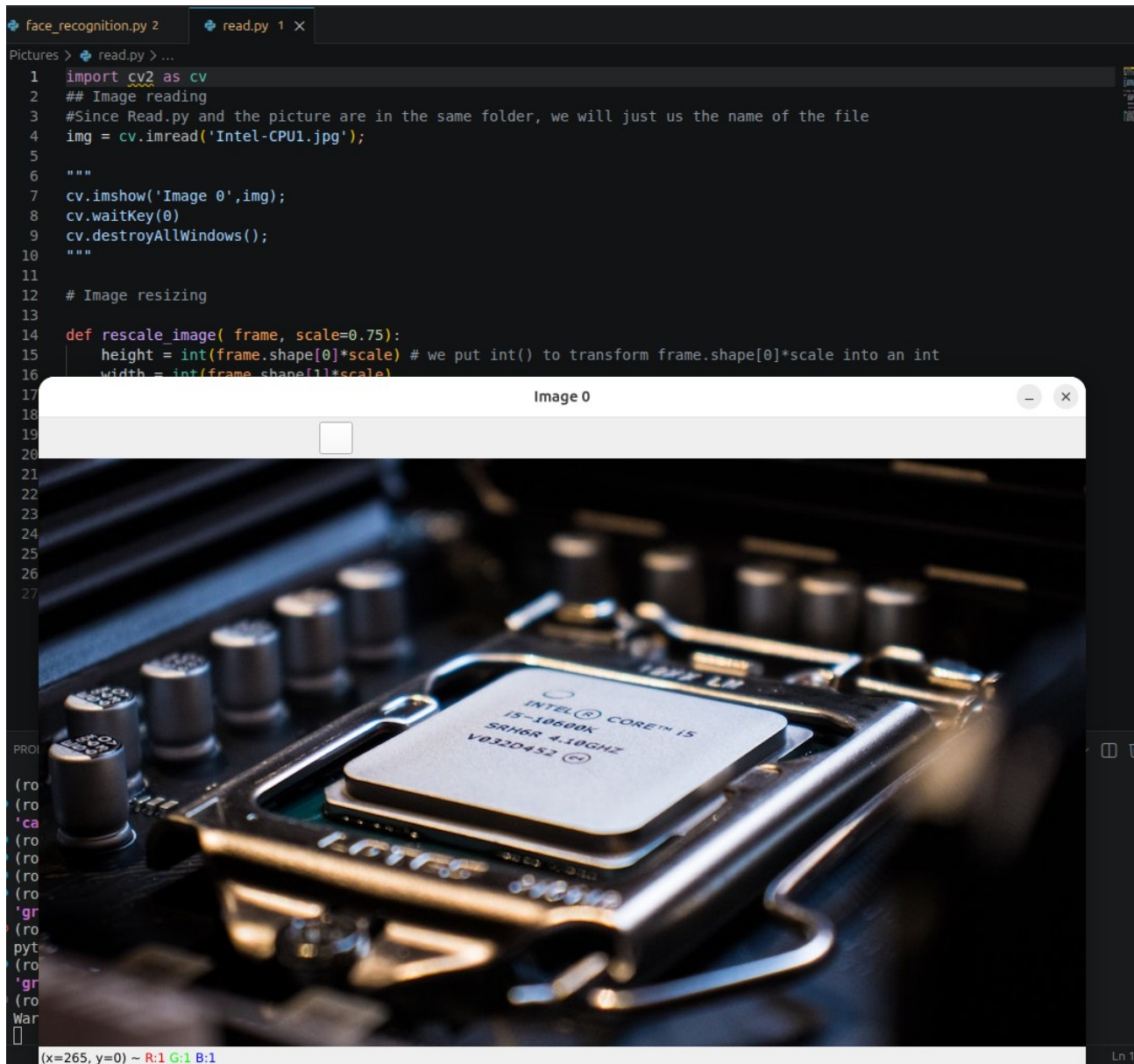
Image reading

```
import cv2 as cv

img = cv.imread('/home/kelly/XDU_internship.github.io/OpenCV_Files/Intel-CPU1.jpeg');
cv.imshow('Image 0',img);

cv.waitKey(0)
cv.destroyAllWindows();
```

Result:



Video reading

```
1  import cv2 as cv
2
3  vd = cv.VideoCapture('dog.mp4');
4
5  while True:
6      isTrue, frame = vd.read()
7      if isTrue:
8          cv.imshow("video 1",frame);
9
10     else:
11         break;
12
13  vc.release()
14  cv.destroyAllWindows()
```

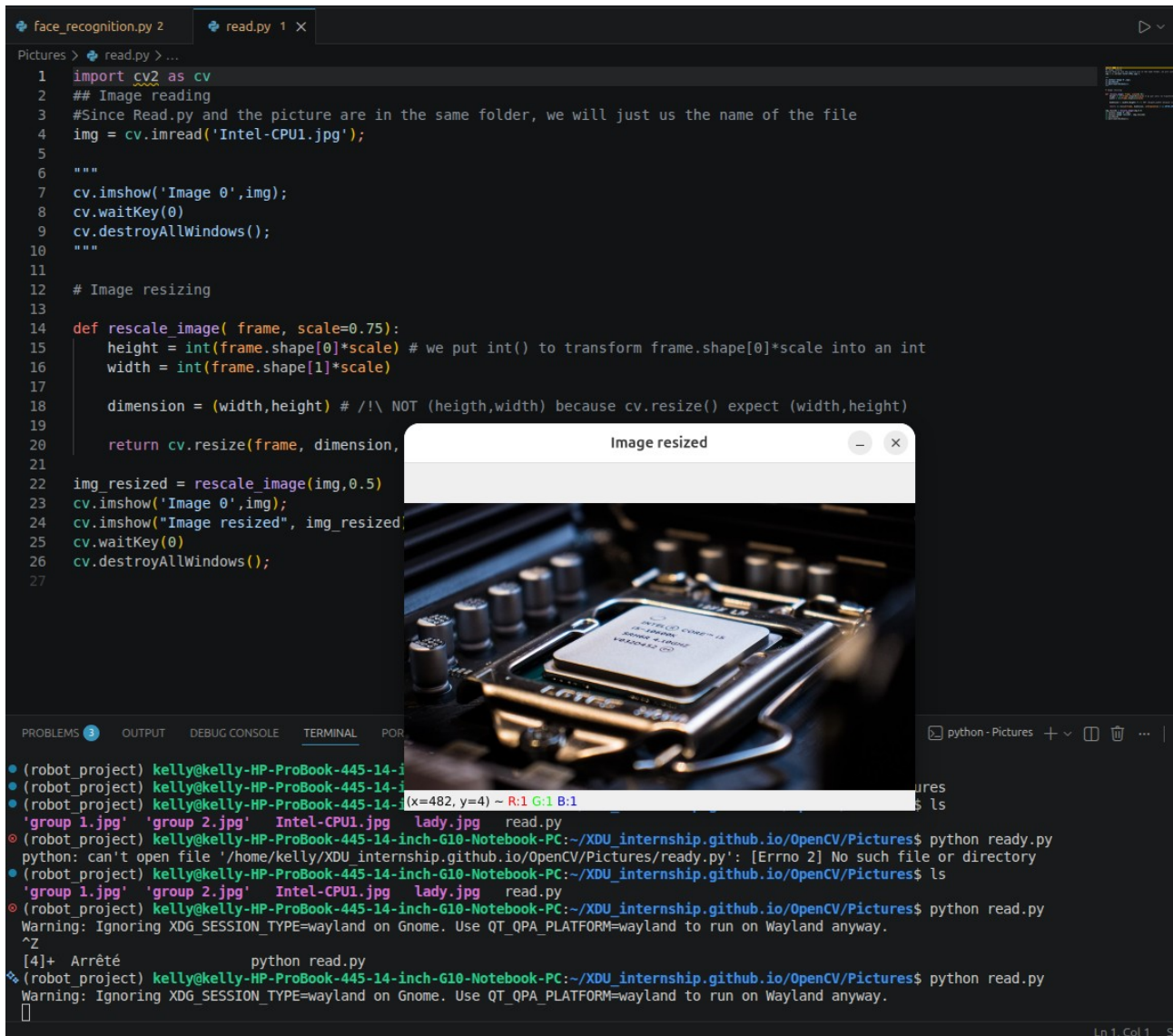
The execution of the code above creates a window that disappears as soon as it appears. We need to manage the time. That is the problem that the following code solves.

```
1  import cv2 as cv
2
3  vd = cv.VideoCapture('dog.mp4');
4
5  while True:
6      isTrue, frame = vd.read()
7      if isTrue:
8          cv.imshow("video 1",frame);
9          #display the frames every 20 ms and get out of the video when the letter 'd' is pressed
10         if cv.waitKey(45) & 0xFF ==ord('d') :
11             break;
12     else:
13         break;
14
15  vc.release()
16  cv.destroyAllWindows()
```

`cv.waitKey(45)` allows the frame to be displayed every 45 ms and returns a 32-bit number where the last 8 bits determine the state of the keyboard. If these last 8 bits are OFF, then we continue to display, but if they are ON, we interrupt the display. We use this characteristic to tell OpenCV that if the letter 'd' is pressed, stop the display; otherwise, continue. That is what is interpreted by **"if `cv.waitKey(45)&0xFF == ord('d')`"**. We can decide to display slowly or rapidly by changing x in `cv.waitKey(x)`.

2- Rescaling and resizing a Window

Rescaling a video (works for image, video and live video)



```
1 import cv2 as cv
2 ## Image reading
3 #Since Read.py and the picture are in the same folder, we will just use the name of the file
4 img = cv.imread('Intel-CPU1.jpg');
5
6 """
7 cv.imshow('Image 0',img);
8 cv.waitKey(0)
9 cv.destroyAllWindows();
10 """
11
12 # Image resizing
13
14 def rescale_image( frame, scale=0.75):
15     height = int(frame.shape[0]*scale) # we put int() to transform frame.shape[0]*scale into an int
16     width = int(frame.shape[1]*scale)
17
18     dimension = (width,height) # /\ NOT (height,width) because cv.resize() expect (width,height)
19
20     return cv.resize(frame, dimension,
21
22 img_resized = rescale_image(img,0.5)
23 cv.imshow('Image 0',img);
24 cv.imshow("Image resized", img_resized);
25 cv.waitKey(0)
26 cv.destroyAllWindows();
27
```

The terminal output shows the execution of the script and the display of the rescaled image:

```
(robot project) kelly@kelly-HP-ProBook-445-14-i
(robot project) kelly@kelly-HP-ProBook-445-14-i
(robot project) kelly@kelly-HP-ProBook-445-14-i (x=482, y=4) ~ R:1 G:1 B:1
'group 1.jpg' 'group 2.jpg' Intel-CPU1.jpg lady.jpg read.py
(robot project) kelly@kelly-HP-ProBook-445-14-inch-G10-Notebook-PC:~/XDU_internship.github.io/OpenCV/Pictures$ python ready.py
python: can't open file '/home/kelly/XDU_internship.github.io/OpenCV/Pictures/ready.py': [Errno 2] No such file or directory
(robot project) kelly@kelly-HP-ProBook-445-14-inch-G10-Notebook-PC:~/XDU_internship.github.io/OpenCV/Pictures$ ls
'group 1.jpg' 'group 2.jpg' Intel-CPU1.jpg lady.jpg read.py
(robot project) kelly@kelly-HP-ProBook-445-14-inch-G10-Notebook-PC:~/XDU_internship.github.io/OpenCV/Pictures$ python read.py
Warning: Ignoring XDG_SESSION_TYPE=wayland on Gnome. Use QT_QPA_PLATFORM=wayland to run on Wayland anyway.
^Z
[4]+ Arrête python read.py
(robot project) kelly@kelly-HP-ProBook-445-14-inch-G10-Notebook-PC:~/XDU_internship.github.io/OpenCV/Pictures$ python read.py
Warning: Ignoring XDG_SESSION_TYPE=wayland on Gnome. Use QT_QPA_PLATFORM=wayland to run on Wayland anyway.
```

Rescaling an image or video allows us to adapt the display code to the abilities of the camera we will use. Here are some reasons why resizing video is important:

- Computational efficiency:

Processing a video frame-by-frame can be computationally expensive. A standard 1080p HD frame contains over 2 million pixels. If we are running a complex algorithm (like face detection or object tracking) on every single pixel, the frame rate will drop significantly.

1- The math: reducing the frame dimension by half (e.g., 1080p to 540p) reduces the total pixel count by 75%.

2- Result: the code runs much faster, often moving from "laggy" to "real-time".

- **Machine Learning constraints:**

Most AI and deep learning models (like YOLO, SSD, or MediaPipe) have a fixed input size. For example, if a model has been trained to see images at 224x224 pixels, it can't process a 1920x1080 video directly. We must then resize the video frames to match the model's architecture before feeding them in.

- **Aspect ratio and normalization:**

When dealing with multiple video sources, they often come in different resolutions, and rescaling the image or video gives the following advantages:

1- Consistency: resizing allows us to standardize our data so the analysis logic remains the same regardless of the source.

2- Normalization: it helps reduce noise and high frequencies that might not be necessary for a specific task, such as detecting the presence of a large vehicle.

Difference between rescaling and resizing

Resizing:

It forces an output image to a certain size where the proportionality between width and height can be changed.

- **Goal:** force an image into a fixed container.

- **Risk:** If the target dimensions don't match the original proportions, the images will look stretched or squashed.

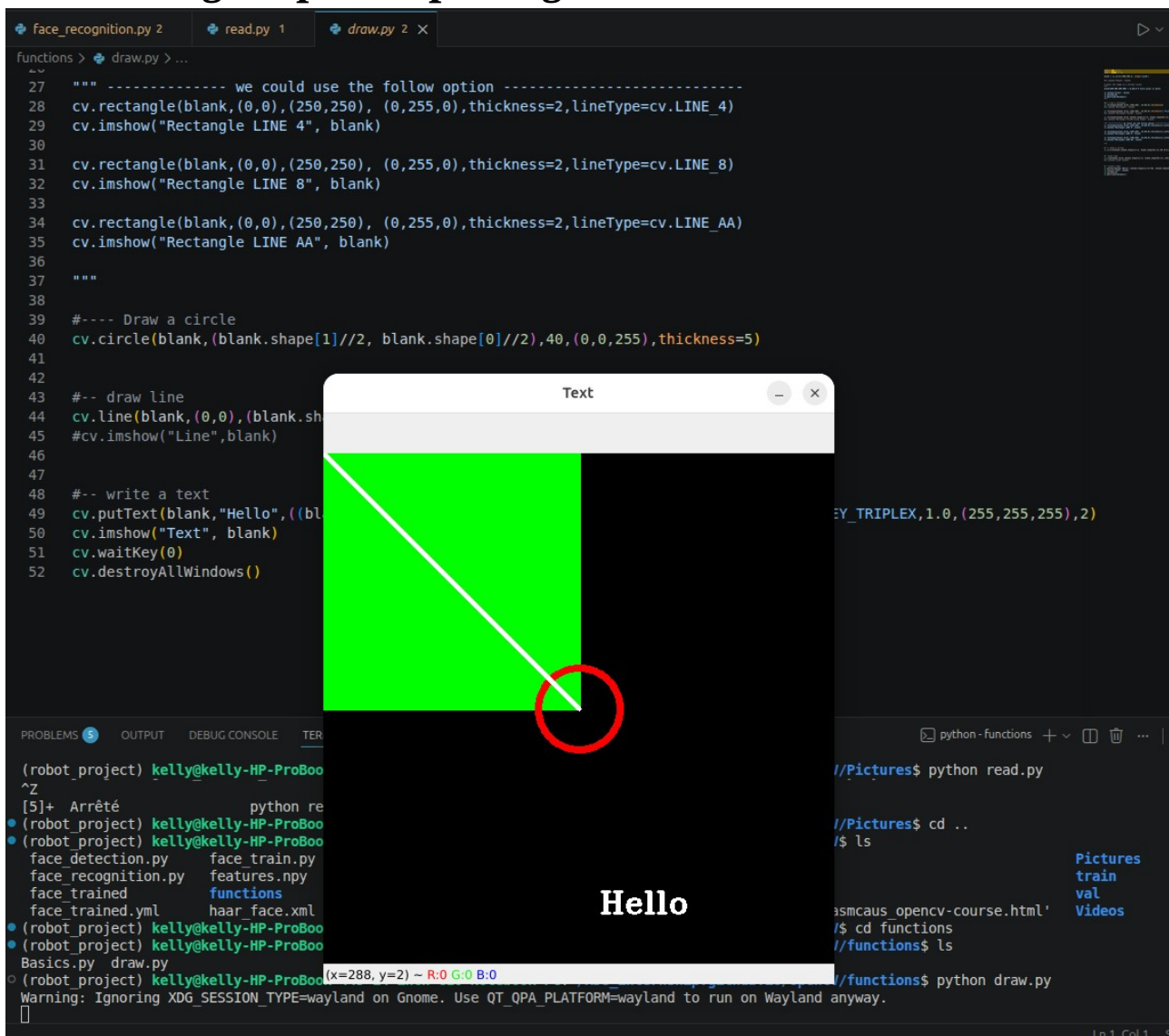
Rescale:

Keeps the proportionality between them. And it is done based on a scale factor or a percentage (e.g., make an image 50% smaller)

Resizing a video (works for only live video)

(refer to code)

3- Drawing shape and putting text



The screenshot shows a code editor with a Python script named `draw.py`. The script uses OpenCV to draw various shapes on a blank image and display them in windows. The code includes comments and function calls for drawing rectangles, a circle, a line, and text. A terminal window at the bottom shows the execution of the script, displaying the output of the `ls` command and the warning message: "Warning: Ignoring XDG_SESSION_TYPE=wayland on Gnome. Use QT_QPA_PLATFORM=wayland to run on Wayland anyway."

```
27 """ ----- we could use the follow option -----
28 cv.rectangle(blank,(0,0),(250,250), (0,255,0),thickness=2,lineType=cv.LINE_4)
29 cv.imshow("Rectangle LINE 4", blank)
30
31 cv.rectangle(blank,(0,0),(250,250), (0,255,0),thickness=2,lineType=cv.LINE_8)
32 cv.imshow("Rectangle LINE 8", blank)
33
34 cv.rectangle(blank,(0,0),(250,250), (0,255,0),thickness=2,lineType=cv.LINE_AA)
35 cv.imshow("Rectangle LINE AA", blank)
36
37 """
38
39 #--- Draw a circle
40 cv.circle(blank,(blank.shape[1]//2, blank.shape[0]//2),40,(0,0,255),thickness=5)
41
42
43 #- draw line
44 cv.line(blank,(0,0),(blank.shape[1],blank.shape[0]),(0,0,255),thickness=2)
45 #cv.imshow("Line",blank)
46
47
48 #- write a text
49 cv.putText(blank,"Hello",((blank.shape[1]//2, blank.shape[0]//2)),cv.FONT_HERSHEY_SIMPLEX,1.0,(255,255,255),2)
50 cv.imshow("Text", blank)
51 cv.waitKey(0)
52 cv.destroyAllWindows()
```

The terminal output shows the execution of the script, displaying the output of the `ls` command and the warning message: "Warning: Ignoring XDG_SESSION_TYPE=wayland on Gnome. Use QT_QPA_PLATFORM=wayland to run on Wayland anyway."

Theses functions will allow us to write the name of the crop recognized on the screen.

In the code **`blank = np.zeros((500,500), dtype='uint8')`**

`-np.zeros()`:

We create an array full of zeros, but it is for initialization.

`-(500,500)`:

means 500 rows and 500 columns, which also means we want 500x500 pixels.

`-dtype='unit8'`:

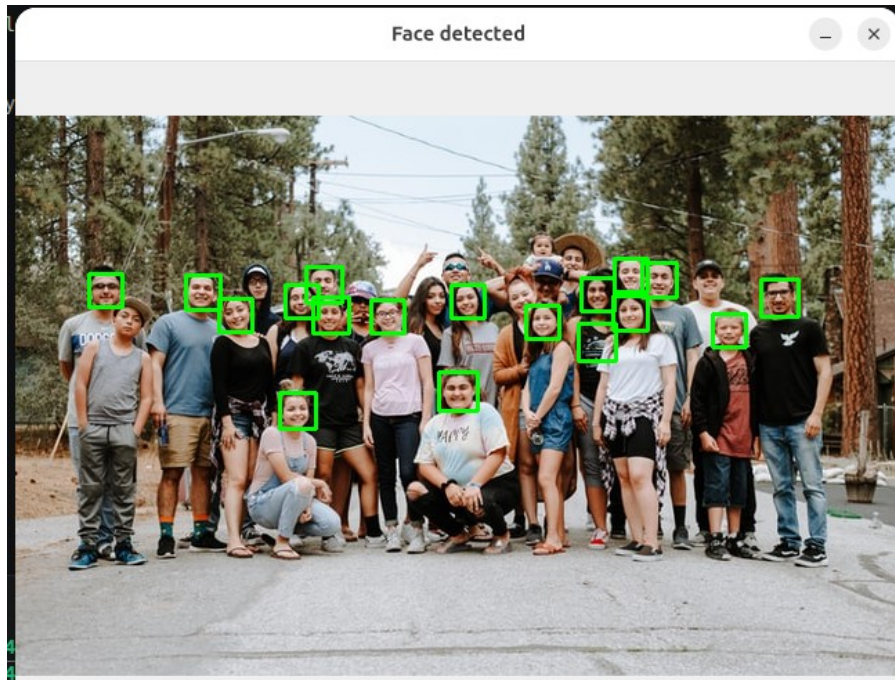
means "unsigned integer, 8 bits," and so each pixel can take 256 (2^8) values from 0 (black) to 255 (white).

4- Face detection

This part is just for human face detection, no matter what the face is.

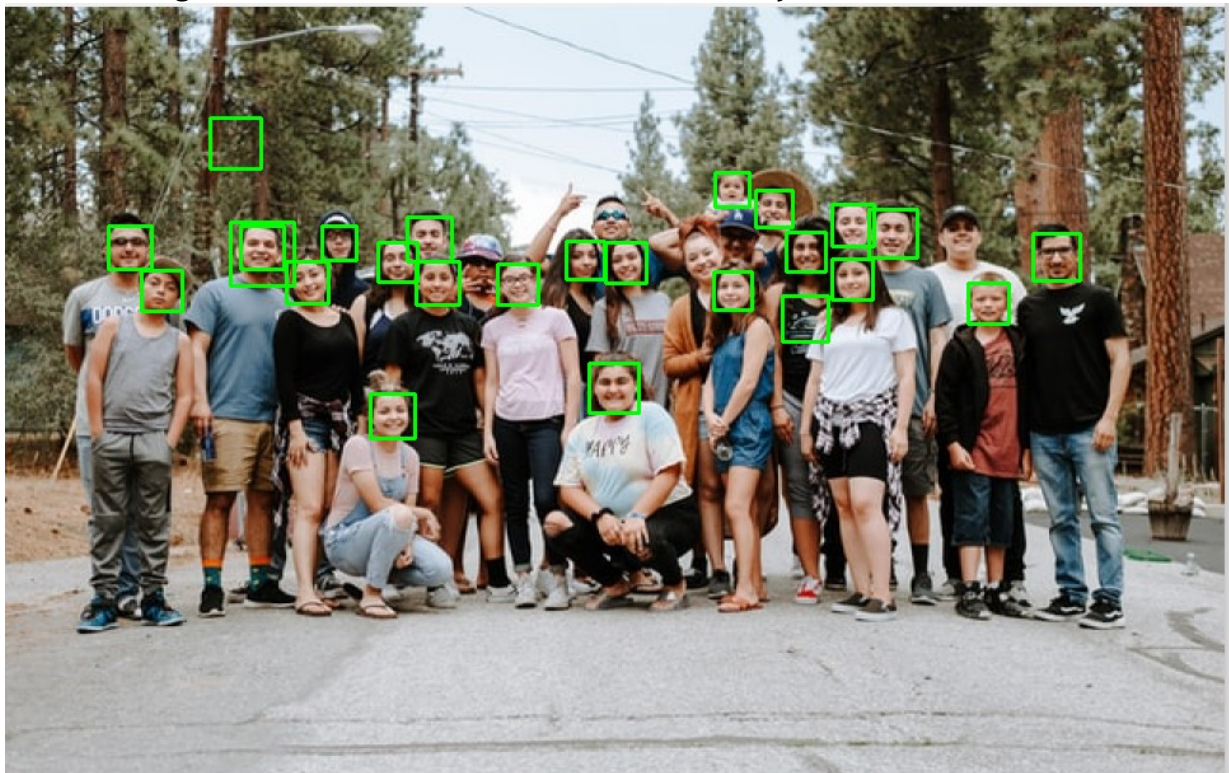
To do this, we will use a classifier, and the one we use here is the “Haar Cascade”.

With the original image, we have the following result:



Efficiency:17/27 faces detected.

With the image resized with the scale=1.5 we have what follows:



By augmenting the size of the image, we have good face detection. But we can also increase the efficiency by changing scaleFactor and minNeighbors.

For : -scaleFactor = 1.25 and minNeighbors =1.

Efficiency:

23/ 27 faces detected.

For: -scaleFactor =1.255 and minNeighbors = 1.

Efficiency:

24/27 face detected.

5- Face recognition

After the face detection, I trained the model for the face recognition, and here is the result.

```
face_recognition.py 2 x
face_recognition.py > ...
1 import cv2 as cv
2 import numpy as np
3
4 haar_cascade = cv.CascadeClassifier('haar_face.xml')
5
6 people = ['Mindy Kaling', 'Elton John', 'Ben Afflek', 'Jerry Seinfeld', 'Madonna']
7
8 face_recognizer = cv.face.LBPHFaceRecognizer_create()
9 face_recognizer.read('face_trained.yml')
10
11 img = cv.imread(r'/home/kelly/XDU_internship.github.io/OpenCV/val/mindy_kaling/3.jpg')
12 gray = cv.cvtColor(img,cv.COLOR_BGR2GRAY)
13
14 #detection of the face
15
16 face_rect = haar_cascade.detectMultiScale(gray, 1.1, 5)
17
18 for (x,y,w,h) in face_rect:
19     face_roi = img[y:y+h, x:x+w, :]
20
21     label, confidence = face_recognizer.predict(face_roi)
22     print(f'Label: {label}, Confidence: {confidence}')
23
24     cv.putText(img, label, (x, y-10), cv.FONT_HERSHEY_SIMPLEX, 1.1, (0, 255, 0), thickness=2)
25     cv.rectangle(img, (x, y), (x+w, y+h), (0, 255, 0), thickness=2)
26
27 cv.imshow("Detected face", img)
28 cv.waitKey(0)
29 cv.destroyAllWindows()
```

